

Proposal: AI-Powered Email Assistant for Fintech Inbox Management

1. What We Want to Build

We receive many emails every day — customer questions, compliance requests, transaction issues, partner updates, and more.

Replying manually takes a lot of time. It also carries risk if someone forgets a compliance rule or sends the wrong tone in a sensitive fintech situation.

Goal: Build a smart AI agent that

- Reads new emails in our inbox
- Learns from all past emails and our previous replies
- Drafts polite, accurate, compliant replies
- Waits for human approval before sending anything
- Keeps tight safety rules

After that, we add a simple dashboard to see stats like how many emails were handled, approval speed, guardrail triggers, etc.

This will save time, reduce mistakes, and keep everything auditable.

2. How It Will Work (High-Level Flow)

1. New email arrives in our inbox (Gmail or Outlook).
2. The system notices it right away (using webhook or polling).
3. It pulls relevant past emails and replies from our secure knowledge base.
4. It sends this context + the new email to Anthropic's latest Claude model.
5. Claude creates a draft reply.
6. Before anything goes out, multiple fintech-specific guardrails check it:
 - No leaking PII
 - No risky financial advice
 - Tone matches our brand
 - Regulatory keywords / phrases are correct
 - No hallucinations or wrong facts
7. If it passes, the draft goes to you for review.
8. You can:
 - Approve → email sends automatically
 - Reject + give feedback → AI improves next time
9. Every action (received, drafted, approved, sent, rejected) is logged in a central database.
10. A dashboard shows real-time stats: emails per day, average response time, approval %, guardrail blocks, etc.

Historical emails are regularly added to the knowledge base so the AI keeps learning from real conversations.

3. Architecture Overview

